

■ 2.6.13 Special Topic: Files

Mathematica provides various ways to use files that are stored on your computer.

The most common use of files in *Mathematica* is as a way of saving definitions that are needed in several different *Mathematica* sessions. The *Mathematica* library contains, for example, files of definitions that can be read into any *Mathematica* session when they are needed.

This section discusses some of the commands in *Mathematica* for reading and writing files. If you use *Mathematica* with a front end, then you may be able to save definitions and input directly using features of the front end, without ever needing to make use of the intrinsic file manipulation capabilities of *Mathematica*.

<code><<file</code>	read in a file
<code>expr >> file</code>	write an expression to a file
<code>expr >>> file</code>	append an expression to a file
<code>!!file</code>	display the contents of a file

Basic input and output operations.

You can use `<<` to read in any file of *Mathematica* input. The result from `<<` is the last expression that was read from the file. (If the last line of the file is blank, then the last expression is taken to be `Null`.) If *Mathematica* finds a syntax error in the file, it reports the error, then stops reading the file.

This reads in the file `Polyhedra.m`, which contains a sequence of *Mathematica* definitions.

```
In[1]:= <<Polyhedra.m
Out[1]= Polyhedra`
```

The `>>` operator is essentially the inverse of the `<<` operator: it allows you to write out *Mathematica* expressions to a file, in a form that you can subsequently read in using the `<<` operator.

This writes the expression `x^2 + y^2` out to the file `tmp.m`.

```
In[2]:= x^2 + y^2 >> tmp.m
```

Here are the contents of `tmp.m`. The expression was written in `InputForm`. Note that `>>` overwrites any previous data in the file.

```
In[3]:= !!tmp.m
x^2 + y^2
```

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Reading in `tmp.m` gives back the original expression.

```
In[3]:= <<tmp.m
```

```
Out[3]= x2 + y2
```

You can put any sequence of *Mathematica* commands into a file, and then read in the file, and execute the commands, using the `<<` operator. It is often convenient to prepare a file of commands outside of *Mathematica*, say using a standard text editor, and then read the file into *Mathematica* when it is ready.

Sometimes you may want to create a complete “script” for all the input to be given in a particular *Mathematica* session. The procedure for running *Mathematica* in this kind of “batch” mode differs from one computer system to another. Section ?? gives some discussion of it.